

Ramireddy Venkata Siva Sandeep

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Hyderabad, Telangana

Professional Summary

- Machine Learning professional with eight plus years of experience in the Data and AI sector, proficient in managing the full lifecycle of data science projects, including exploratory data analysis, feature engineering, model development and deployment.
- Skilled in building multi-agent GenAI systems using Google ADK, enabling intelligent orchestration, automated workflows, and scalable data integration across enterprise platforms.
- Experienced in designing, implementing, and iterating on generative AI models such as GPT, Gemini models with a focus on Retrieval-Augmented Generation to improve model responses and data retrieval.
- Proficient in optimizing models for coherence, accuracy, minimal hallucination, and maximum coverage.
- Successfully migrated Oozie models to GCP Cloud Composer, leading to cost reductions and streamlined deployment. Contributed to GCP modernization by optimizing BigQuery table storage.
- Designed, optimized, and deployed machine learning models to predict customer transaction values, driving better customer targeting and business outcomes.

Professional Skills

Programming: Python, PySpark, SQL

ML & AI: Supervised/Unsupervised ML, Deep Learning, NLP, Time Series (ARIMA, Prophet), Clustering, Feature Engineering.

Generative AI: Gemini models(nano banana& Veo 3.1) ,GPT, Llama, Mistral, Falcon, Prompt Engineering, RAG, LLM Fine-Tuning, LangChain, Multi-Agent Systems, google ADK

Big Data & Pipelines: Spark, Dataproc, BigQuery, Airflow/Cloud Composer, ETL Pipelines

Cloud & DevOps: GCP, Azure, Jenkins, Git/GitHub, DevOps

Tools: VS Code, Anaconda, Jira, pandasAI, Google ADK

Professional Experience

06/2017 – Present, (8 years 6 months)

Senior Machine Learning

Tiger Analytics • Hyderabad, Telangana (Jan 2025 - current)

Project: Agentic GenAI Content Automation for Yum! Brands (Taco Bell).

Developed an agentic GenAI content editing platform to streamline reuse and adaptation of marketing images and videos across global markets. Architected a multi-agent system using ADK with dedicated agents for image editing, video transformation, content interpretation, and workflow coordination. Automated asset customization, resizing, brand compliance checks, and multi-market adaptation. Integrated prompt-driven decision logic for contextual content generation, improving creative throughput and production efficiency. Enabled scalable, consistent, and automated campaign content generation.

Project: Multi-Agent Data Integration Framework for GE Appliances.

Designed and developed a multi-agent, GenAI-powered data integration solution to unify enterprise data sources such as SharePoint and BigQuery. Built dynamic orchestration workflows using the Agent Development Kit (ADK) with specialized agents for querying, data retrieval, transformation, and validation. Ensured scalability, modular design, and secure cross-platform data access. Delivered a fully automated, extensible agent ecosystem that improved data accessibility and reduced manual intervention.

Senior Data Scientist

INCEDO INC • Hyderabad, Telangana (Nov 2022 - Jan 2025)

Project Description: Developed and enhanced a Incedo's Lighthouse platform by integrating machine learning features to improve data analysis and prediction. This included implementing a Key Driver Analyzer using Spearman correlation and Random Forest for feature importance assessment. I also built a Platform Cluster Analyzer utilizing K-means, hierarchical clustering, and DBSCAN techniques to identify patterns in the data. Additionally, I deployed time series models such as ARIMA and Prophet for accurate anomaly detection and forecasting. These enhancements significantly advanced the platform's ability to provide actionable insights and proactive decision-making. Clean and organize data to improve comprehension and processing by the models.

Data Analyst

MAVERIC Systems • Bangalore, Karnataka (April 2022 - Nov 2022)

Project Description: Led the productionizing and monitoring of machine learning models on the GCP platform, meeting the client's requirement to deploy models using ephemeral clusters within Cloud Composer. I gathered requirements from the business team and set up GCP environments for development, preproduction, and production. I orchestrated ML pipelines by creating Airflow DAGs and managing Dataproc clusters. Additionally, I implemented Spark jobs for training large-scale models, configuring Spark properties, and managing deployment cycles. I also scheduled and monitored the deployed ML Spark jobs to ensure smooth and efficient execution.

Senior ML Data Associate

Amazon • Hyderabad, Telangana (June 2017 - April 2022)

Project Description: Developed machine learning models to predict customer transaction values using financial behavior data. Performed extensive data profiling, validation, and feature engineering using Python and Pandas. Built predictive models including Linear Regression, Logistic Regression, Random Forest, and XGBoost for accurate forecasting. Applied PCA and feature selection techniques to optimize model performance. Contributed to CI/CD deployment workflows and supported end-to-end model monitoring. Led team activities including data preparation, model experimentation, reporting, mentoring junior analysts, and performance improvement initiatives. Delivered accurate transaction value predictions that supported improved customer targeting and business decision-making.

Education

Diploma (AI and ML)

04/2023, University of Hyderabad. • Hyderabad, Telangana

BTech Mechanical

06/2017, Lovely Professional University • Jalandhar, Punjab

Certificates

Google Cloud Certified Professional Machine Learning Engineer, (DP-900): Azure Data Fundamentals (DP-100): Azure Data Scientist Associate, ML Practitioner Dataiku